

Vela founder prediction - concept note

Updated follow-on project proposal: Stability-aware integration of LLM-derived feature families

1. Motivation from the current results

The most promising follow-on project treats the current failure mode as a stability and integration problem, not simply a feature-generation problem. The key result so far is that LLM-engineered features improved on the human baseline, whereas reasoning-derived features often increased validation performance without delivering comparable private-test gains. That suggests the main challenge is not whether these richer features contain signal at all, but whether correlated, low-support feature families can be admitted without overfitting under hidden-test shift.

A closely related question concerns deterministic LLM-engineering at larger scale. If LLMs can rapidly generate hundreds or thousands of candidate structured features, then the project should ask how those feature banks ought to be filtered, grouped, reduced, and admitted into the model so that holdout performance improves without losing interpretability.

Proposed paper thesis. LLM-derived feature sets can be highly redundant and deceptively strong in validation. In founder-success prediction, the main challenge is not only generating richer features, but controlling correlated, low-support feature families so that performance survives true holdout evaluation.

2. Main project: stability-aware integration of LLM-derived feature families

The project should ask whether correlated feature families can be integrated in a way that preserves true holdout performance rather than merely boosting cross-validation. This framing should explicitly cover both the current reasoning-derived families and a larger bank of deterministic LLM-engineered features.

- When do LLM-engineered and reasoning-derived feature families genuinely add out-of-sample value?
- Why do some reasoning families improve cross-validation but fail to generalise?
- How should a large deterministic LLM-engineered feature bank be filtered, grouped, and reduced so that it remains interpretable and improves holdout performance?
- Which integration strategy best preserves holdout precision, calibration, and top-k ranking stability while remaining interpretable?

Recommended pipeline

Stage	What to do
1. Baseline and scope	Use the existing strongest structured baseline as the anchor for comparison, then evaluate how

	additional LLM-derived feature families behave relative to it.
2. Family organisation	Organise features into families. Initially this can be human / deterministic LLM-engineered / reasoning A-F. If the deterministic LLM-engineered bank is large, create sub-families or clusters within it and treat those as the modelling unit.
3. Family diagnostics	For each family, quantify redundancy and incremental value using correlation or cluster overlap, conditional gain over the baseline, fold-to-fold stability, calibration, and precision@k impact.
4. Controlled admission	Add families sequentially rather than concatenating everything at once. Retain a family only if it improves inner-CV performance and remains stable across folds or resamples.
5. Small model ladder	Keep the comparison hypothesis-led: (i) elastic net baseline, (ii) one grouped or stability-aware route such as stability selection or sparse-group lasso, and (iii) one compression route such as PLS or sparse PLS for clearly high-collinearity families. Use naive concatenation only as a negative control.
6. Residual sub-analysis	Test whether added families explain residual error or improve ranking among borderline cases. This gives a direct test of orthogonal value rather than raw score inflation.
7. Decision rule	Prefer the route that best reduces the CV-to-holdout gap while keeping the final model interpretable and stable, even if it is not the most complex.

3. Final outputs

- One final comparison table showing the structured baseline and 2-3 selected stability-aware variants, with validation and holdout metrics.
- One compact diagnostic figure showing where feature families helped, where they destabilised the model, and how the CV-to-holdout gap changed.
- A short recommendation on whether any reasoning family adds robust orthogonal value beyond the baseline, including the role of residual modelling.
- One interpretable reduced representation for the deterministic LLM-engineered feature bank, for example 10-30 retained features, a small number of grouped summaries, or a compact set of components with clear labels.

Ideal outcome. A convincing result would be either (i) a stability-aware route that matches or improves holdout precision while reducing validation-to-holdout collapse, or (ii) a clear null result showing that many added features mainly inflate validation unless they are tightly filtered.

4. Extension and fallback framing

Structured-first + semantic grouping extension using an LLM

This is the main stretch extension. Ask an LLM to map features into semantic groups, for example trajectory, ownership, coherence, or risk/upside, and compare those groups against prompt-source groups, manual groups, and correlation-based clusters. Keep this as an extension unless it clearly beats simpler grouping strategies.

Fallback paper framing if method gains are weak

If the methodological gains are mixed, the project can still support a strong benchmark/protocol paper focused on evaluation under imbalance and hidden-test shift. In that version, the emphasis would shift toward threshold selection, metric choice (F0.5 / PR-AUC / precision@k), calibration, leakage traps, repeated-split robustness, and the tendency of richer feature families to inflate validation more than true holdout performance.

5. Suggested reading

Theme	Paper	Why it helps
Stability	Meinshausen & Buhlmann (2010), Stability Selection	Canonical reference for subsampling-based stability rather than one lucky split.
Grouped shrinkage	Simon, Friedman, Hastie & Tibshirani (2013), A Sparse-Group Lasso	Natural reference when feature families are the modelling unit.
Compression	Chun & Keles (2010), Sparse Partial Least Squares Regression for Simultaneous Dimension Reduction and Variable Selection	Useful when feature banks are dense, correlated, and still potentially predictive.
Correlated features	Kamkar et al. (2016), Stabilizing l1-norm Prediction Models by Supervised Feature Grouping	Directly relevant to learning or refining groups among correlated predictors.
LLM-guided selection	Jeong, Lipton & Ravikumar (2025), LLM-Select: Feature Selection with Large Language Models	Useful background if the semantic-grouping extension is attempted.
LLM-guided regularisation	Zhang et al. (2025), LLM-Lasso: A Robust Framework for Domain-Informed Feature Selection and	Optional reference on LLM-informed sparse modelling and how to guard against overtrusting

	Regularization	the LLM.
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Practical recommendation. Keep the project tightly scoped and compare a small number of stability-aware routes for admitting extra families. Treat semantic grouping with an LLM as the main extension.